CFD Study of Compressible Flow in a Converging Nozzle

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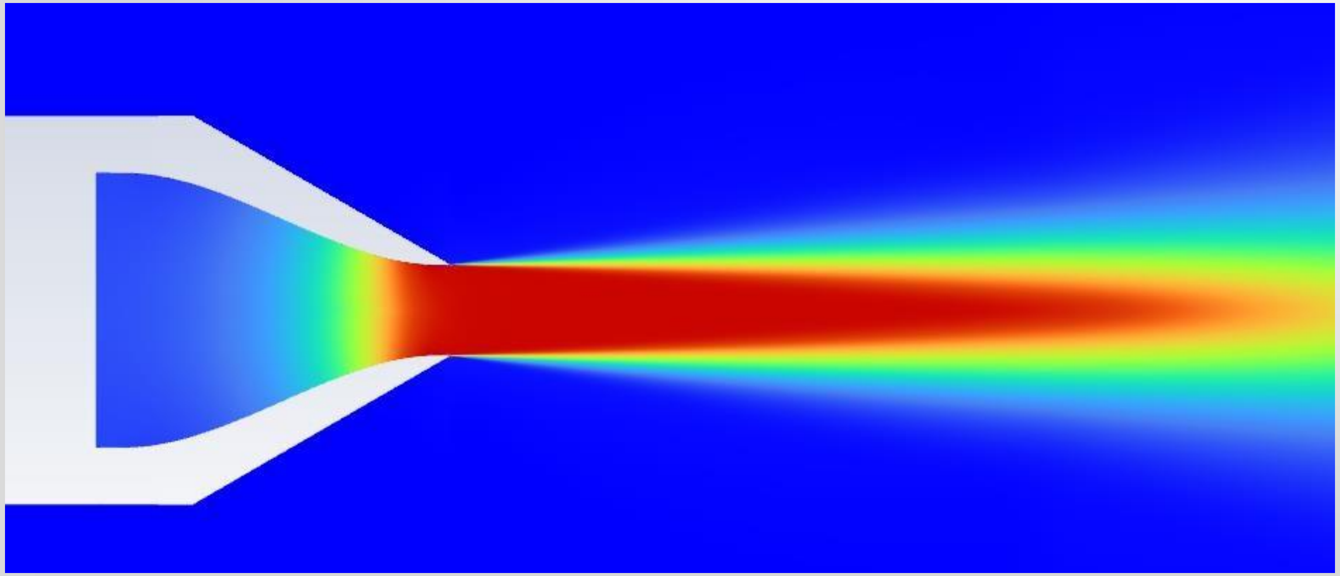
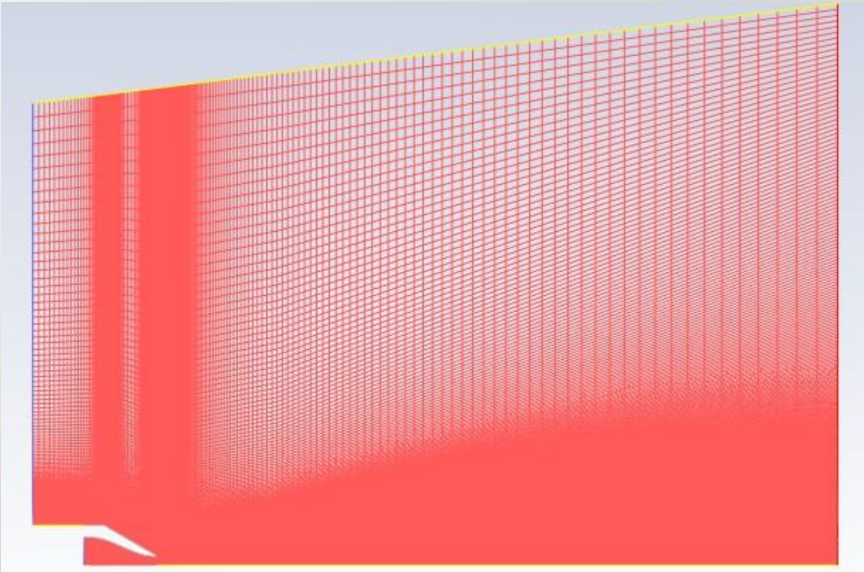




Introduction

Objectives: I perform a comprehensive numerical simulation of compressible subsonic flow in a converging nozzle, validated against experimental measurements and NASA CFD results.

Thrust Evaluation: The thrust generated by the nozzle is computationally evaluated and validated against theoretical limits.

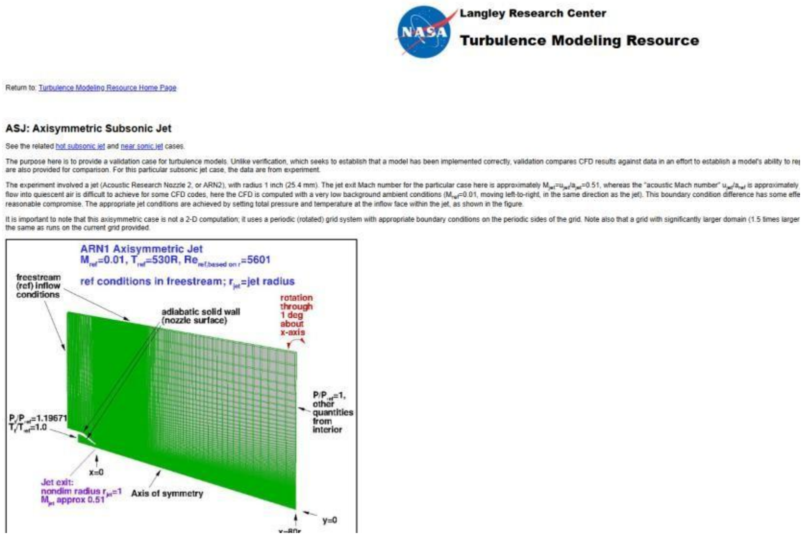
Numerical Framework: The computational workflow utilizes ANSYS DesignModeler for geometry construction, ANSYS Meshing for structured grid generation, and ANSYS Fluent as the finite volume solver.

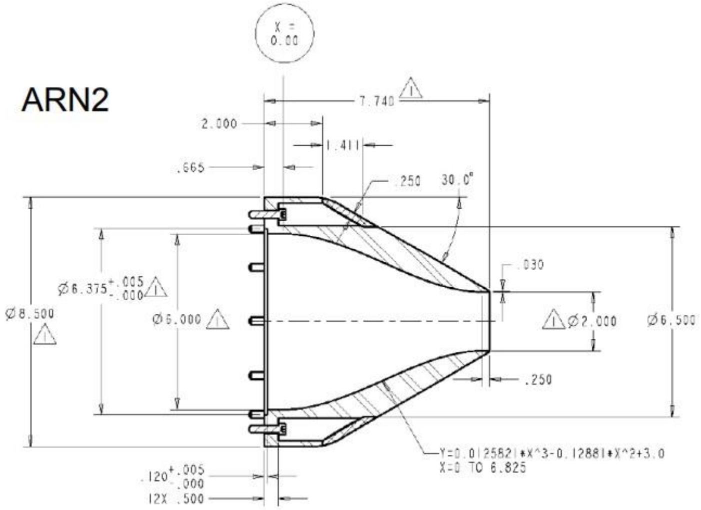
Validation Parameters: Key aerodynamic parameters, specifically the centerline axial velocity and radial velocity distributions, are compared to assess model fidelity.

Introduction

Reference Case: This validation study focuses on the ARN2 (Axisymmetric Radio Nozzle 2) geometry, a benchmark case obtained from the NASA Turbulence Modeling Resource (TMR).

Geometric Specifications: The detailed geometry profile and coordinate specifications are derived from NASA Technical Reports (NASA/TM-2005-213799).





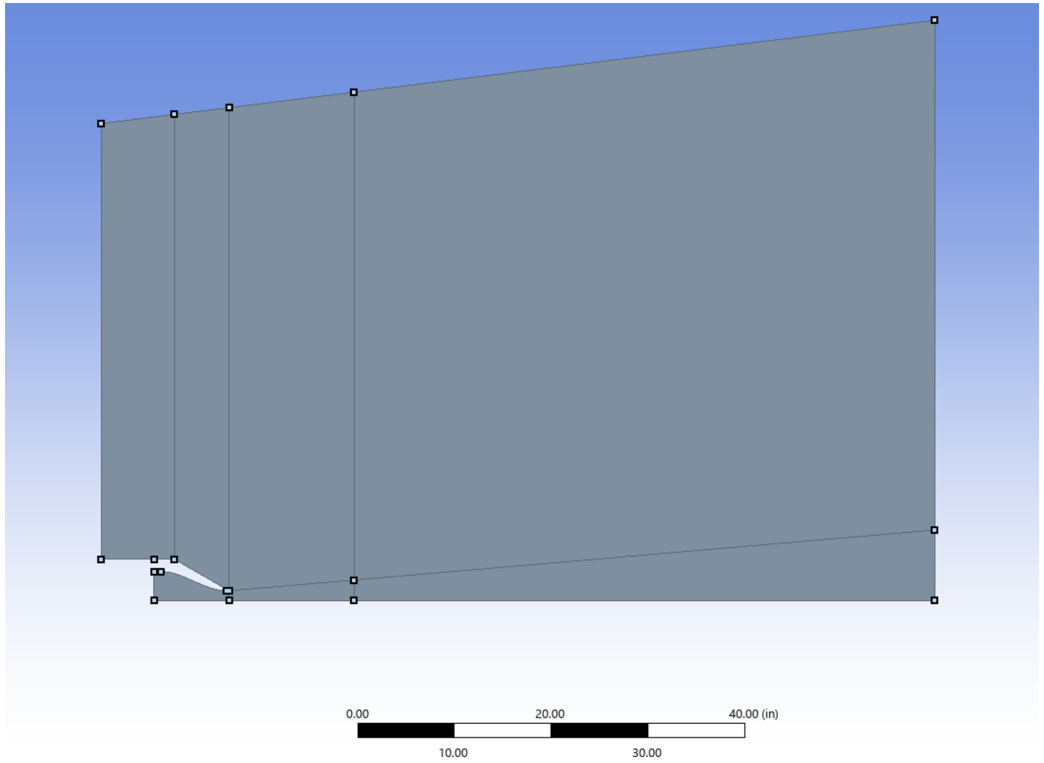
Geometry

Geometry Derivation: The nozzle inner contour is modeled from x = 0 to x = 6.825 inches using reference coordinate polynomials.

$$Y = 0.0125821X^3 - 0.1288X^2 + 3.0 \quad \text{for } X = 0 \text{ to } 6.825$$

Domain Boundaries: The computational domain spans from -12.5 inches upstream of the exit to 80.0 inches downstream, extending 50.0 inches radially at the inlet and 60.0 inches at the outlet, representing a free-jet configuration.

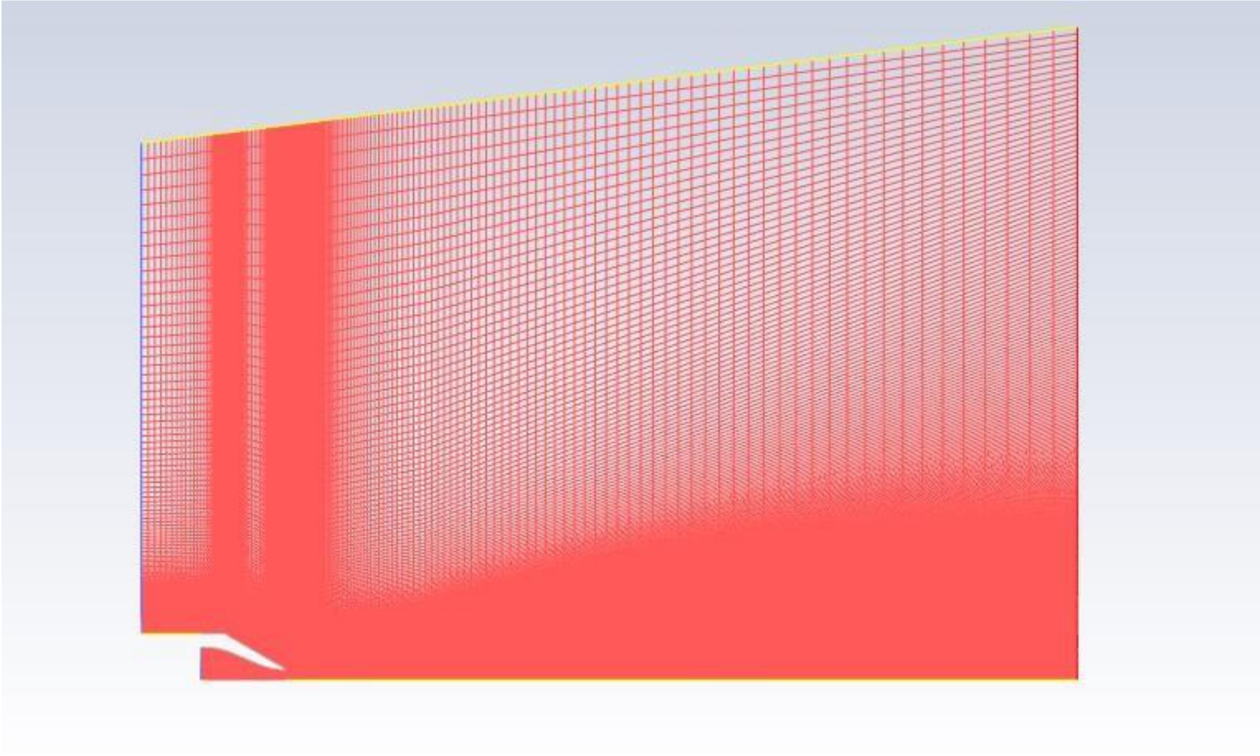
Grid Alignment: The domain is structured to match the NASA-standard validation grids.



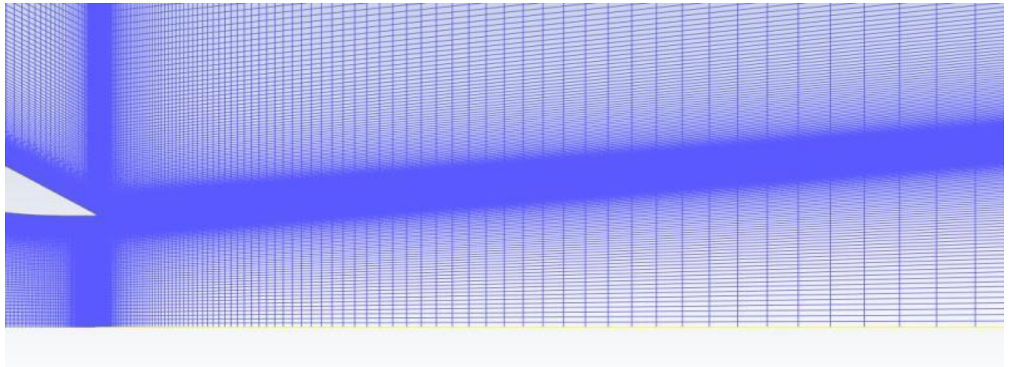
Mesh for ARN2 Nozzle

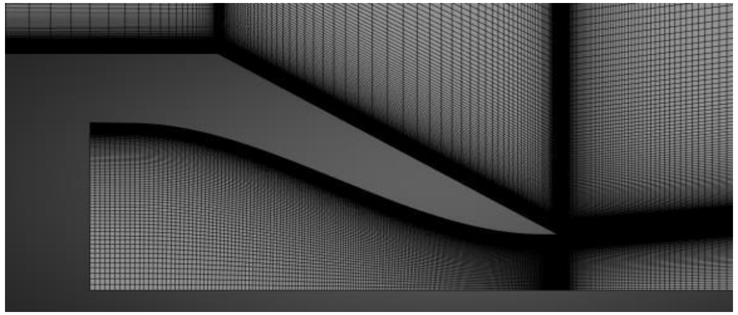
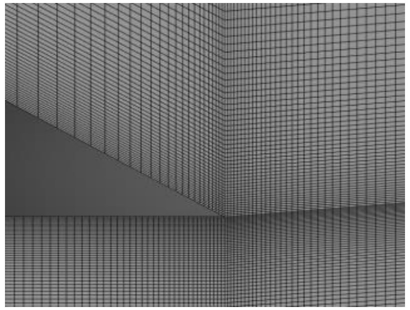
Grid Generation: Structured grids are generated using ANSYS Meshing, resolving the boundary layer with a wall-normal grid spacing yielding y+ < 5.

Mesh Quality: The baseline grid comprises approximately 100,000 hexahedral elements with a minimum orthogonal quality of 0.17, meeting numerical stability criteria.

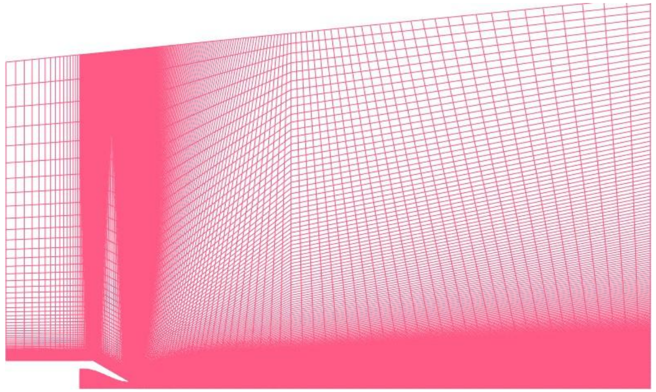


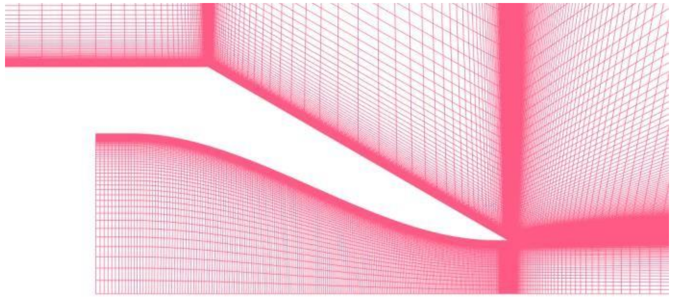
Mesh for ARN2 Nozzle



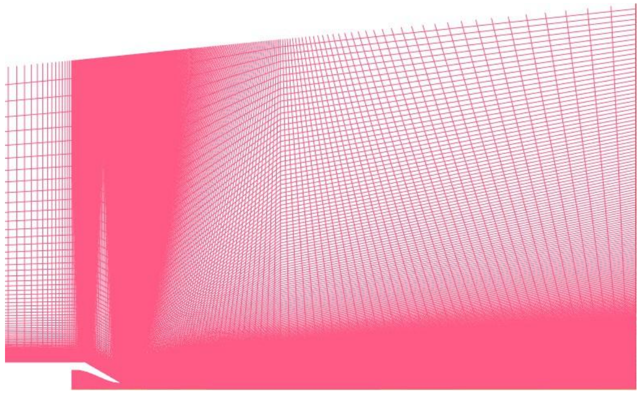



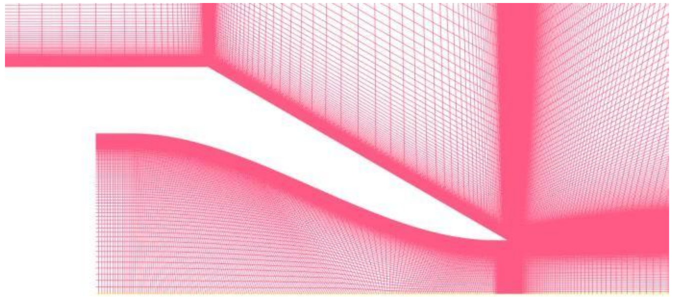
Three Meshes for Mesh independence study



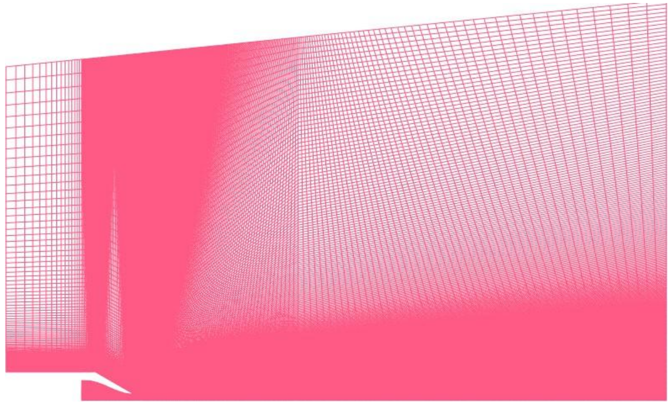


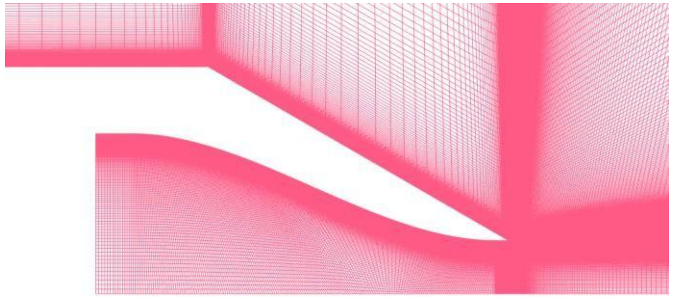
70,600 cells
(Coarse Grid)





128,740 cells
(Medium Grid)

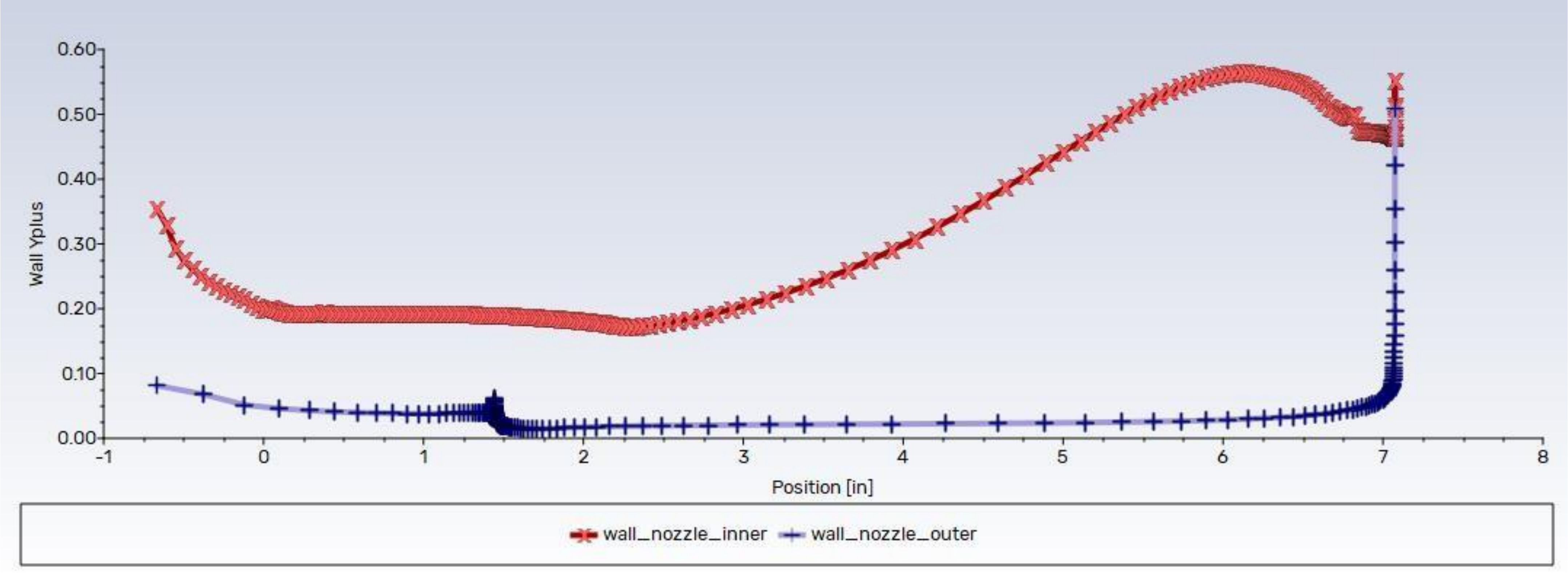




214,900 cells
(Fine Grid)

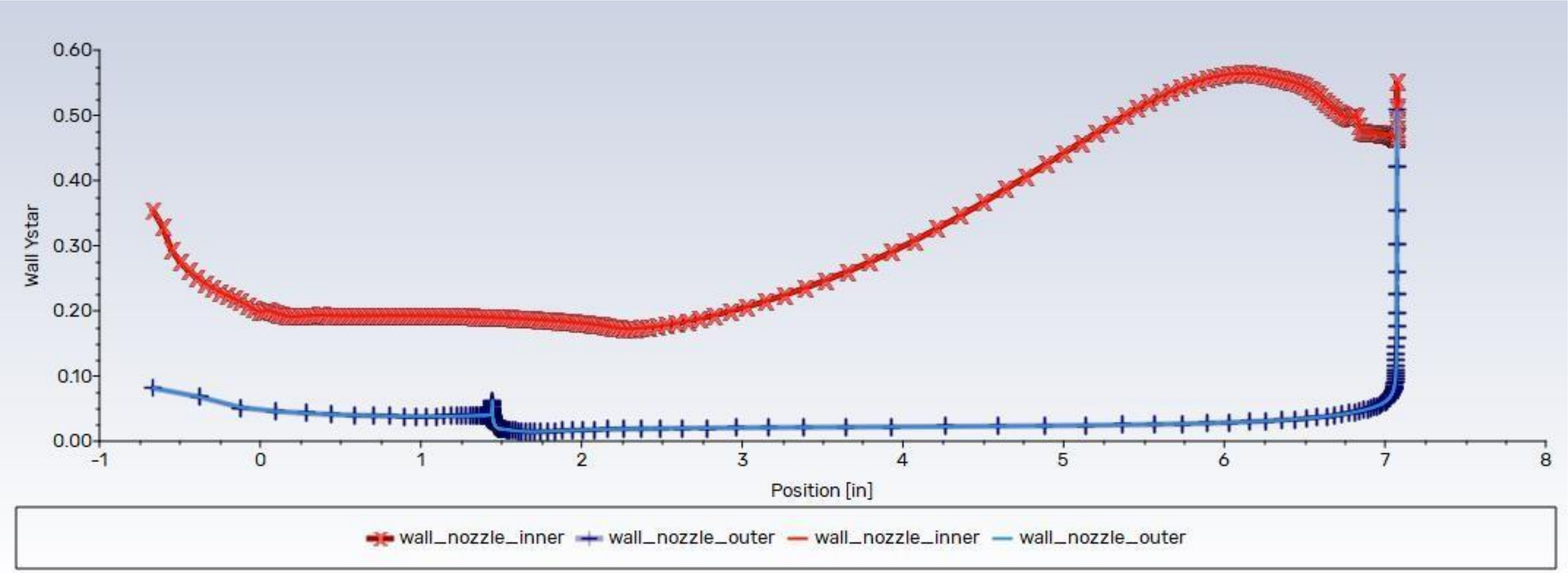
Y+ distribution for the Fine mesh SST model

Boundary Layer Resolution: Wall y+ values are maintained below 1.0 across the entire nozzle wall boundary for all three grid levels under the SST k-ω model, ensuring direct resolution of the viscous sublayer without wall functions.

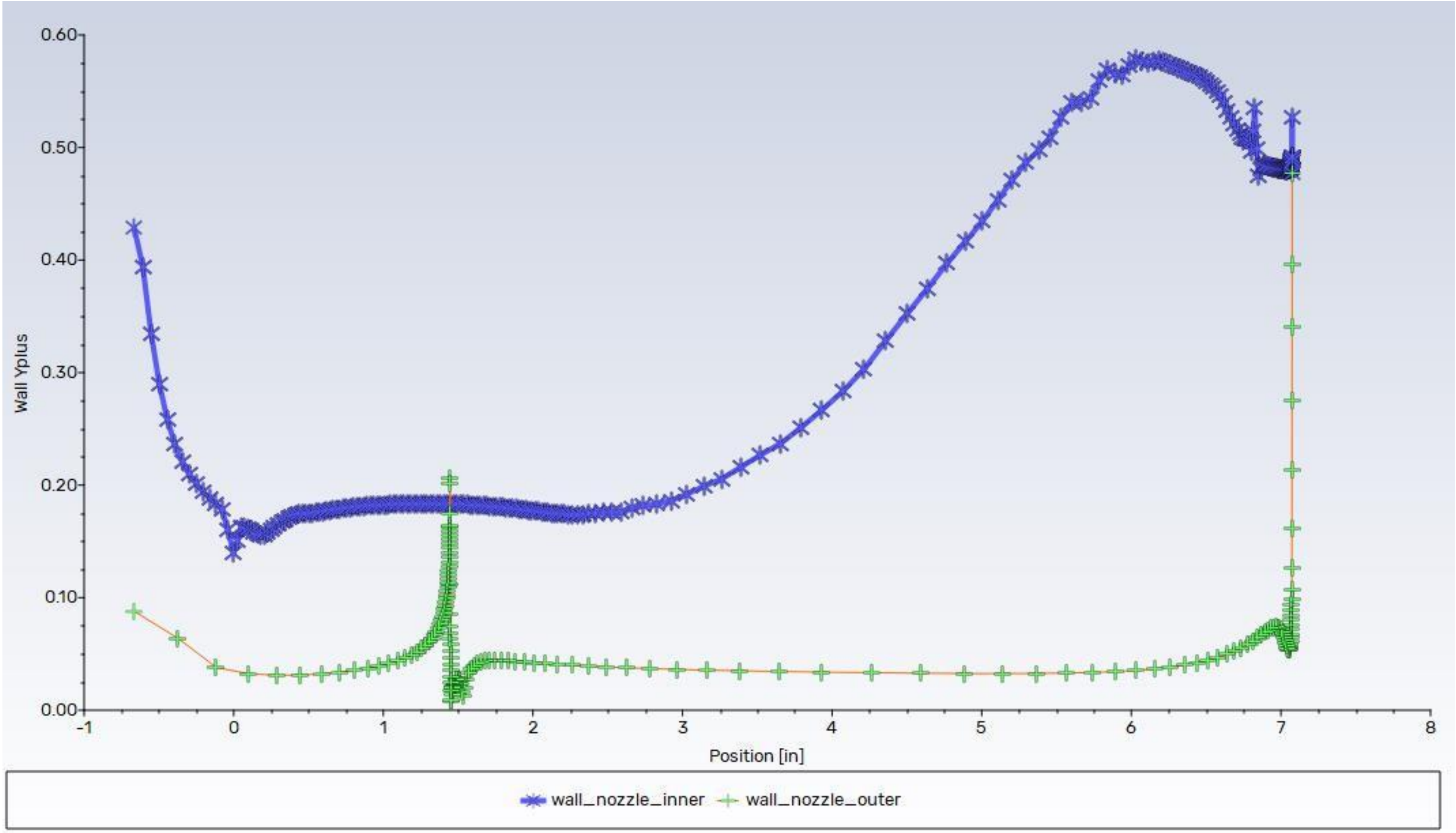


Y+ vs Y* distribution for the Fine mesh SST model

Grid Resolution Equivalence: Comparison of y+ and y* distributions shows negligible differences, verifying the suitability of the boundary layer spacing for low-Reynolds-number modeling.



Y+ distribution for the Fine mesh SA model



Important parameters

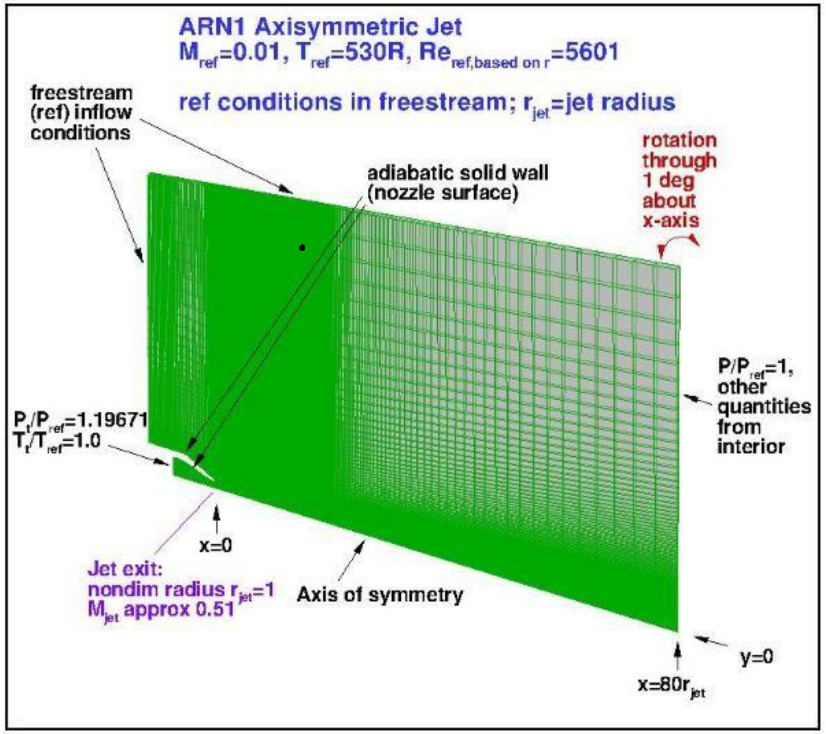
Reference Length: Nozzle throat radius (R_ref) = 1.0 inch (25.4 mm).

Flow Parameters: Reynolds number (Re_D) = 5,601 based on reference throat diameter.

Thermodynamic States: Reference temperature T_ref = 294.4 K (530 R), Reference pressure P_ref = 101,325 Pa.

Total Conditions: Nozzle inlet total-to-reference pressure ratio P_t/P_ref = 1.19671 (P_t = 121,256.6 Pa); total temperature T_t = 294.4 K.

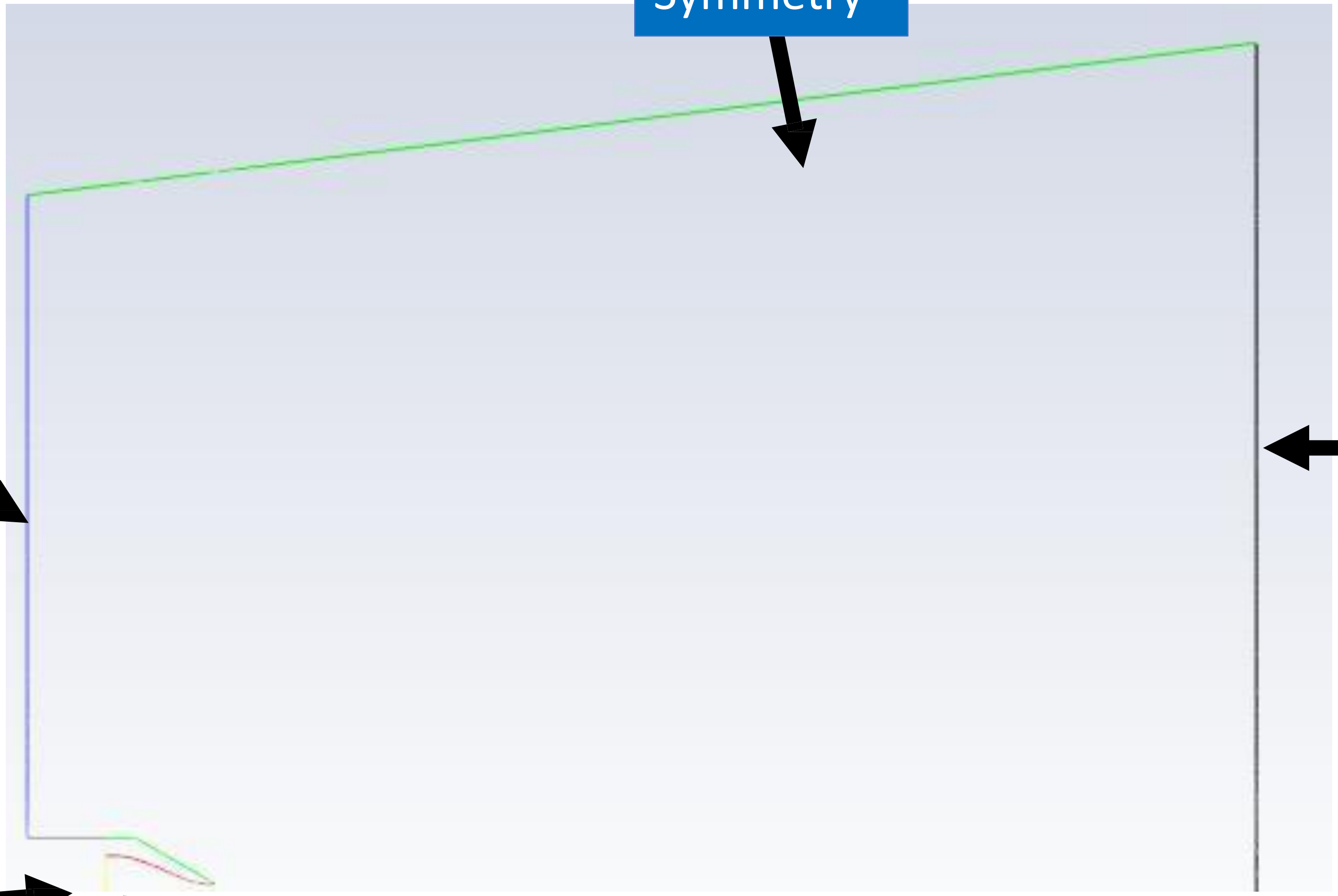
Jet Exit State: Jet Mach number (M_jet) = 0.51.



Boundary Conditions

Operating pressure = 0 Pa

Symmetry



Pressure inlet
P_{total} = 101325 Pa
P_{static} = 101325 Pa
T_{total} = 294.4 K

Pressure Outlet
101325 Pa
294.4 K

Pressure inlet
P_{total} = 121,256.64 Pa
P_{static} = 101325 Pa
T_{total} = 294.4 K

Axis

Boundary Conditions Configuration

Freestream Inlet: Pressure inlet boundary condition configured with total pressure Pt = 101,325 Pa, static pressure Ps = 101,325 Pa, and total temperature Tt = 294.4 K.

Nozzle Inlet: Pressure inlet boundary condition with total pressure Pt = 121,256.64 Pa, static pressure Ps = 101,325 Pa, and total temperature Tt = 294.4 K.

Outlet Boundary: Pressure outlet boundary condition set to a static pressure of 101,325 Pa and backflow temperature of 294.4 K.

Symmetry & Centerline: Axisymmetric boundary conditions enforced along the centerline axis (y = 0); symmetry boundaries defined at domain outer bounds.

Operating Pressure: Set to 0 Pa to ensure absolute pressure calculations across the density-based coupled solver.

Computational Settings

Solver Configuration: A density-based coupled solver is employed to resolve the compressible flow equations.

Turbulence Modeling: Turbulence closure is achieved using the Spalart-Allmaras (SA) and SST k-ω models to compare their performance in shear-layer development.

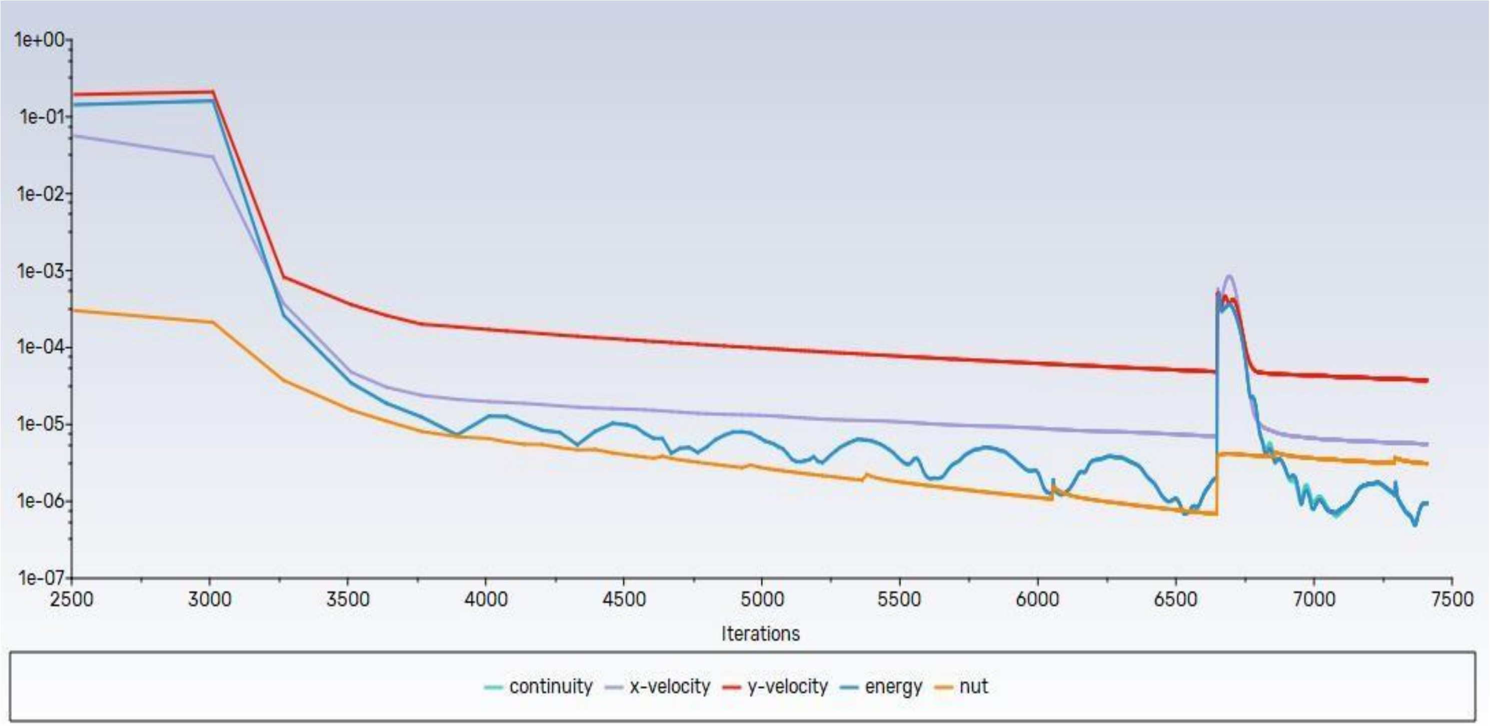
Spatial Discretization: A second-order upwind scheme is utilized for the momentum equations to ensure spatial accuracy.

Fluid Modeling: Air is modeled as an ideal gas, with dynamic viscosity computed via Sutherland's law to account for temperature dependencies.

Initialization & Convergence: I implemented Full Multi-Grid (FMG) initialization and solution steering, with the Courant-Friedrichs-Lewy (CFL) number scaled from 0.1 to 100 over 5,000 iterations to achieve steady-state convergence.

Solution Convergence

- **Residual Convergence History**



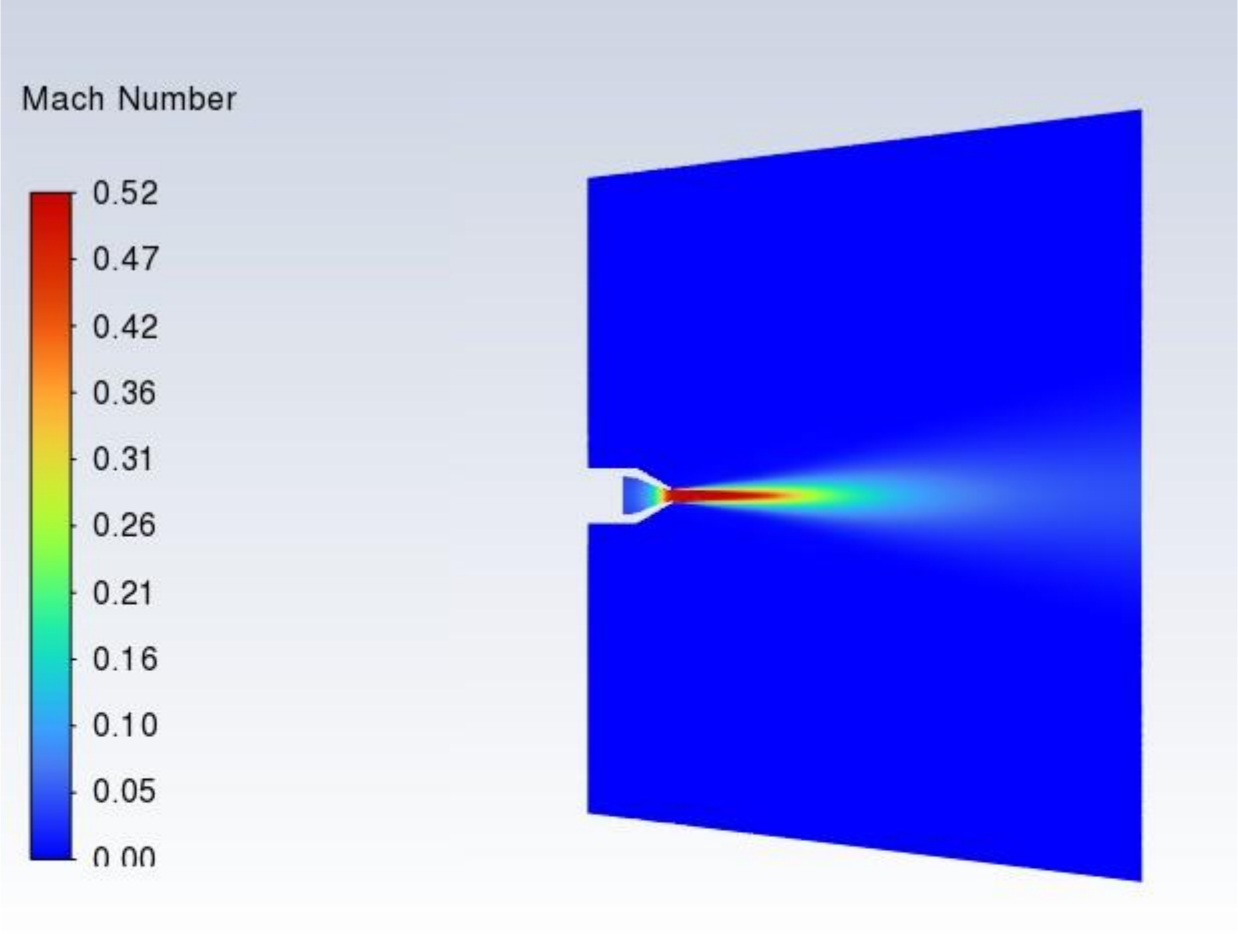
Solution Convergence Analysis

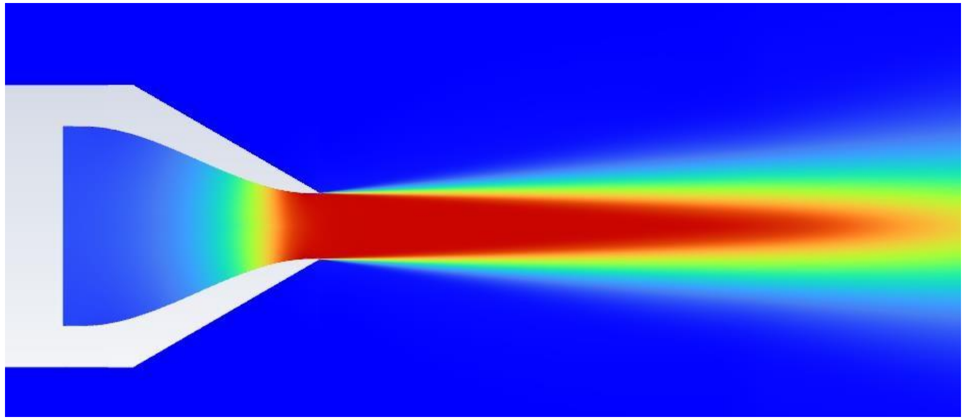
Residual Monitoring: Solution convergence is evaluated based on normalized residual behavior, monitored to a strict threshold of < 10^-6 for all equations.

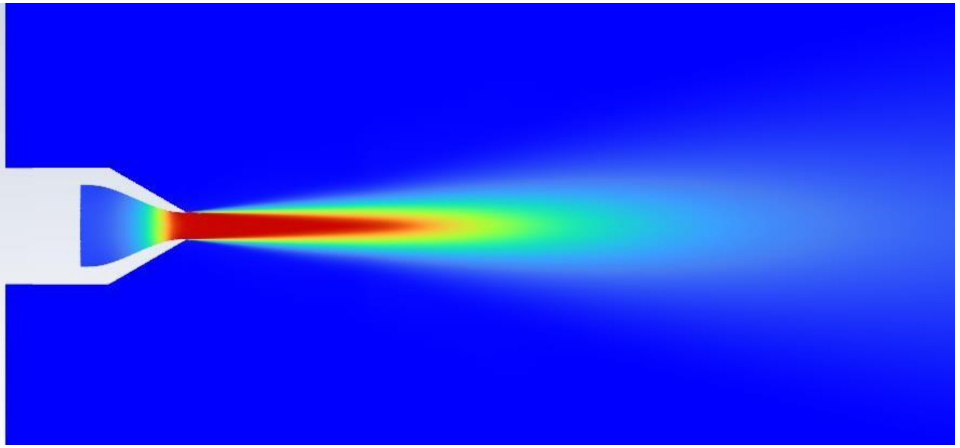
Conservation Balances: Mass and energy flow rates across the nozzle inlets and outlet are verified to ensure conservation properties.

Asymptotic Stability: Centerline axial velocity and wall shear stress profiles are monitored to guarantee asymptotic convergence over 5,000 iterations.

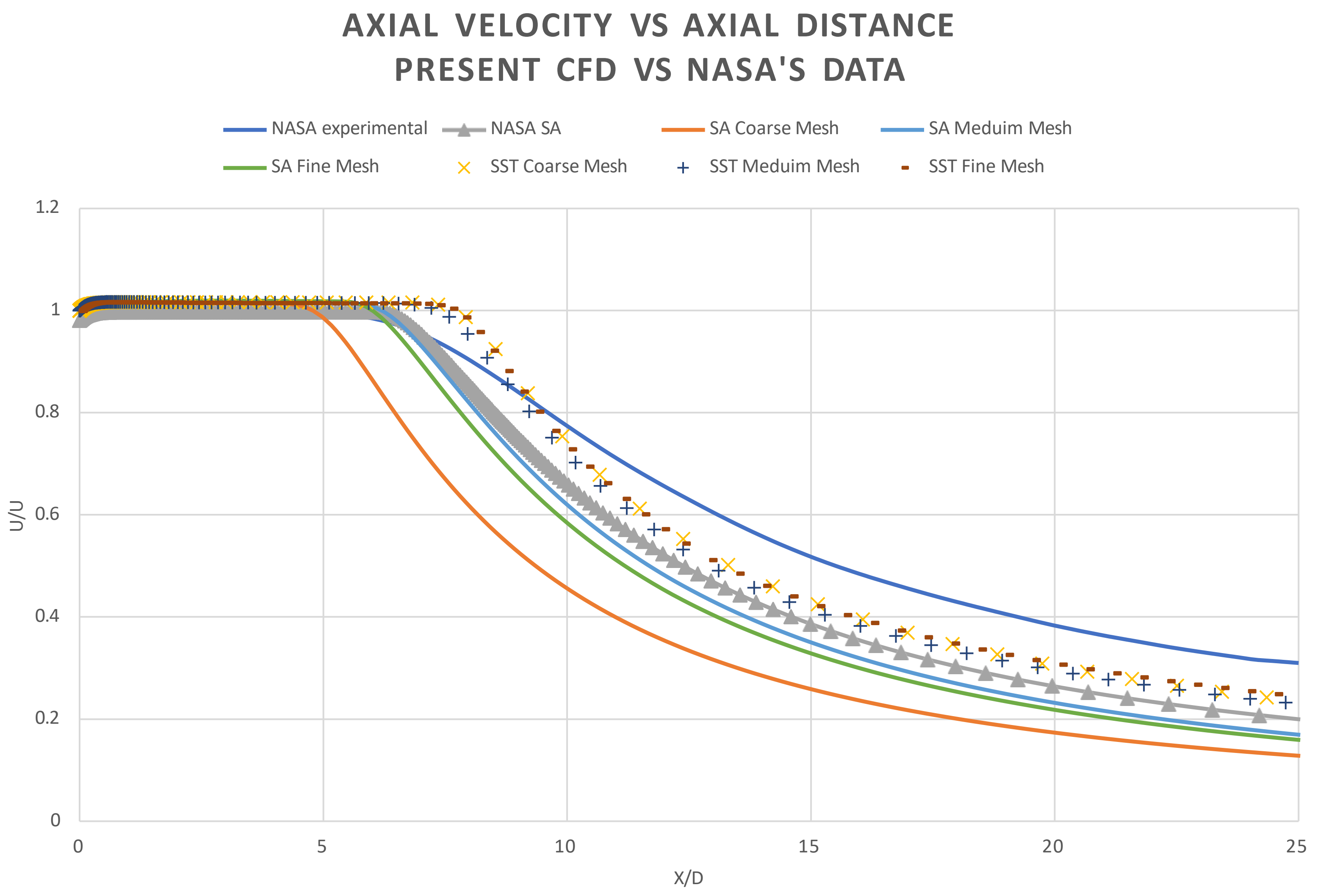
Results – Mach No. Contours
SST Coarse Mesh Y + ~ 1







Validation – Velocity along centerline



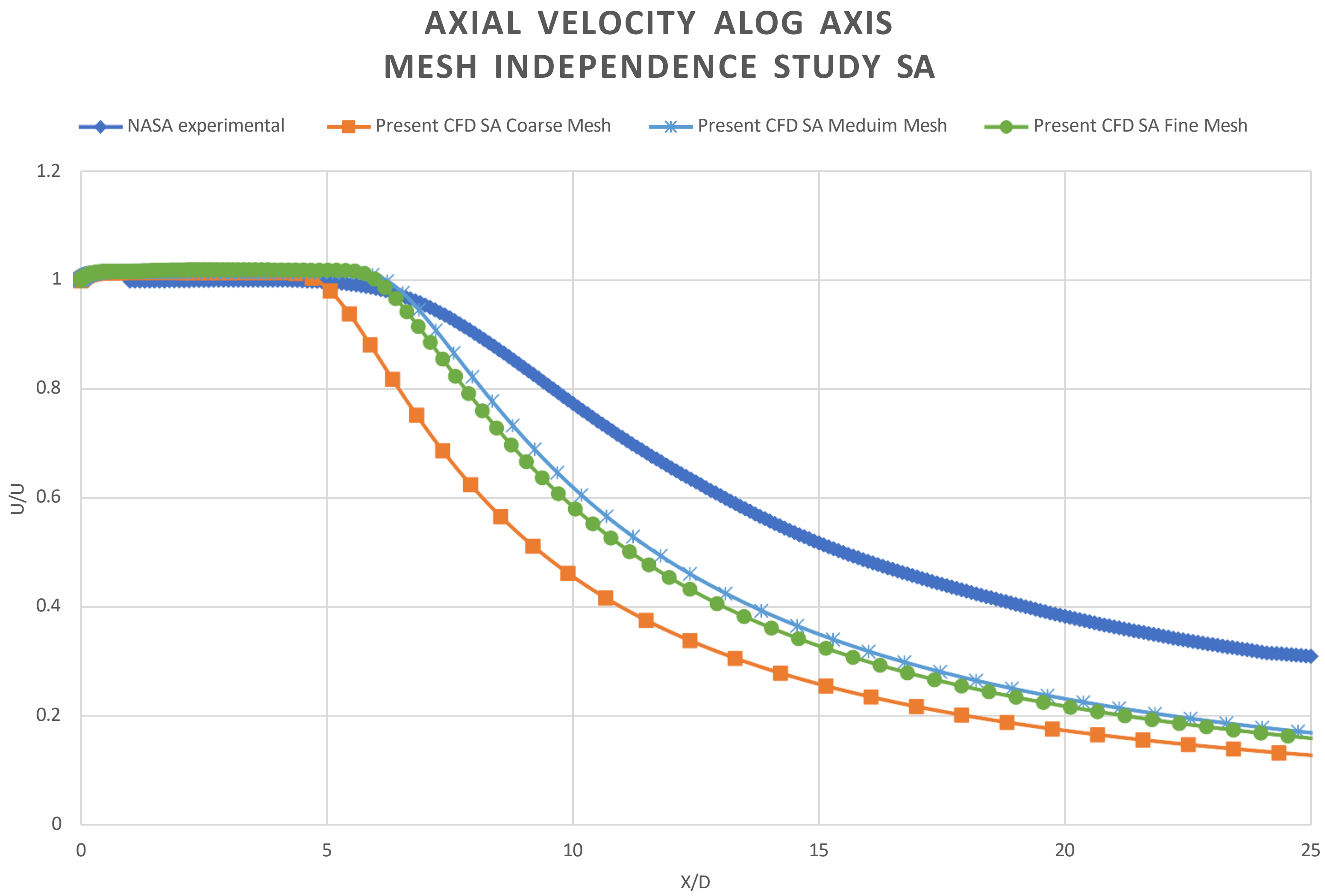
Centerline Velocity Validation Overview

Validation Parameter: The centerline axial velocity ratio (U/Uj) is plotted against normalized axial distance (X/Dj) from the nozzle exit to evaluate solver accuracy.

Validation Datasets: CFD centerline profile predictions for Spalart-Allmaras (SA) and SST k-ω models are validated against NASA subsonic jet experimental data.

Near-Exit Agreement: All configurations of mesh resolution and turbulence models show highly consistent, correct behavior within the jet potential core region ($X/D_j < 5.0$).

Validation – Velocity along centerline



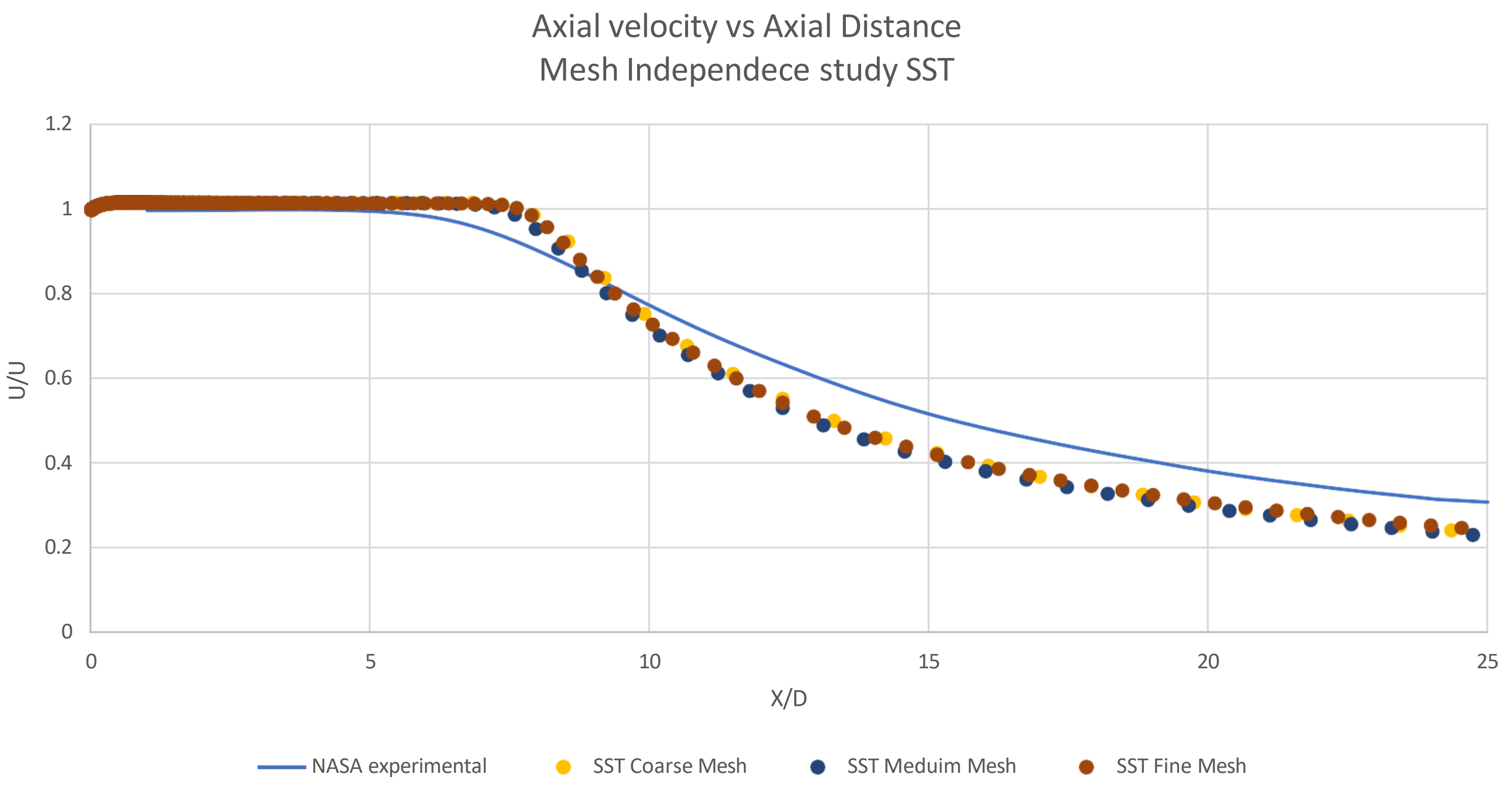
Spalart-Allmaras (SA) Grid Convergence

Grid Density Sensitivity: The Spalart-Allmaras (SA) model demonstrates noticeable grid sensitivity, especially in predicting jet dissipation downstream (X/Dj > 10.0).

Coarse Mesh Performance: The Coarse mesh (70.6k cells) underpredicts jet dissipation downstream, showing an average deviation of 5.13% relative to the Fine mesh.

Mesh Convergence: The Medium mesh (128.7k cells) achieves solid grid convergence, tracking the Fine mesh (214.9k cells) profile within a 1.53% average difference.

Validation – Velocity along centerline



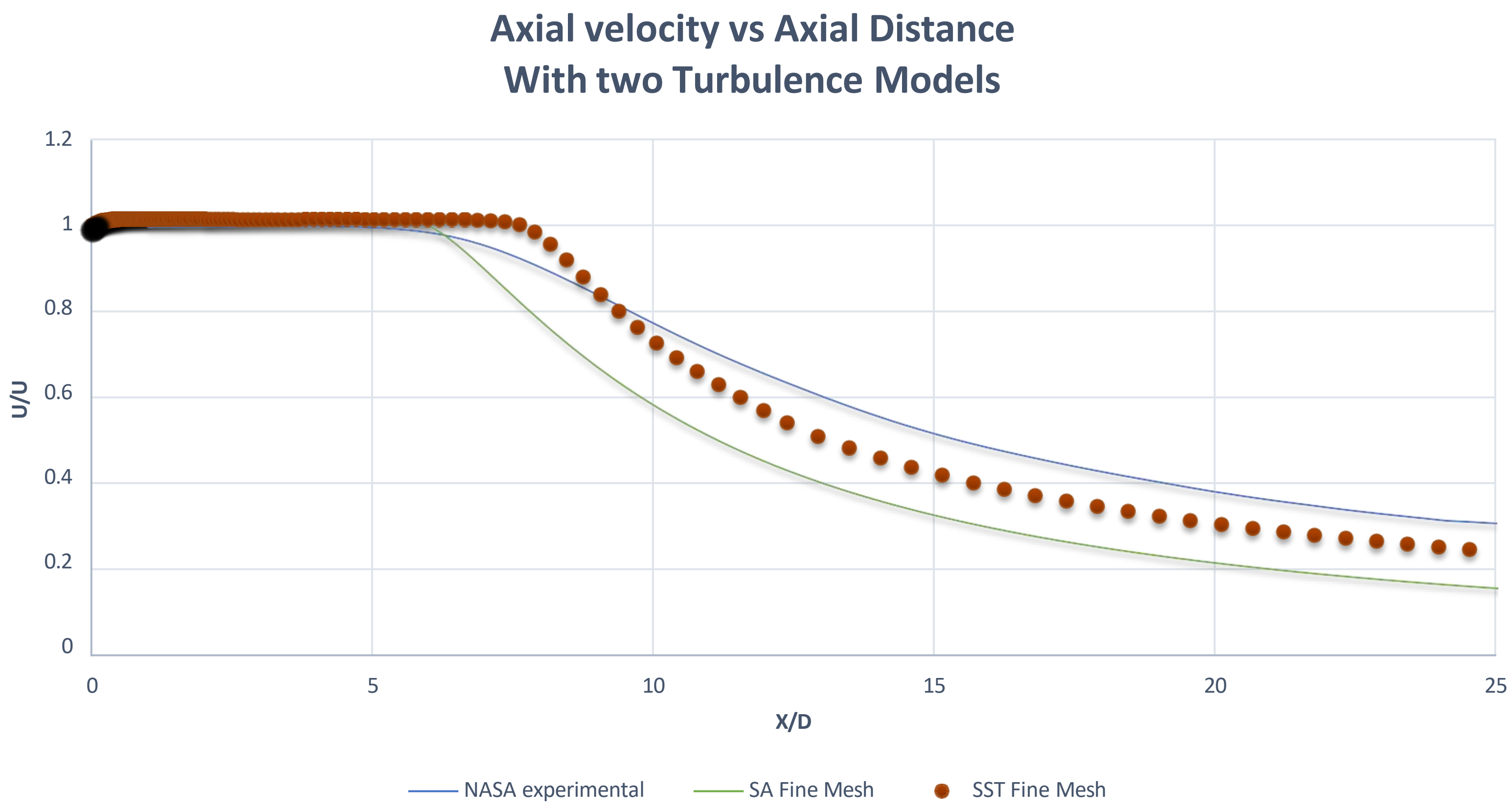
SST k-ω Grid Convergence

Grid Density Robustness: The SST k-ω model shows remarkable numerical stability, showing minimal sensitivity to grid refinement along the centerline axis.

Coarse Grid Validation: The Coarse grid (70.6k cells) provides excellent predictions, deviating by only 0.73% on average from the Fine grid (214.9k cells).

Grid Independence: The Medium grid (128.7k cells) matches the Fine grid profile within 1.01% average difference, verifying grid independence.

Validation – Velocity along centerline



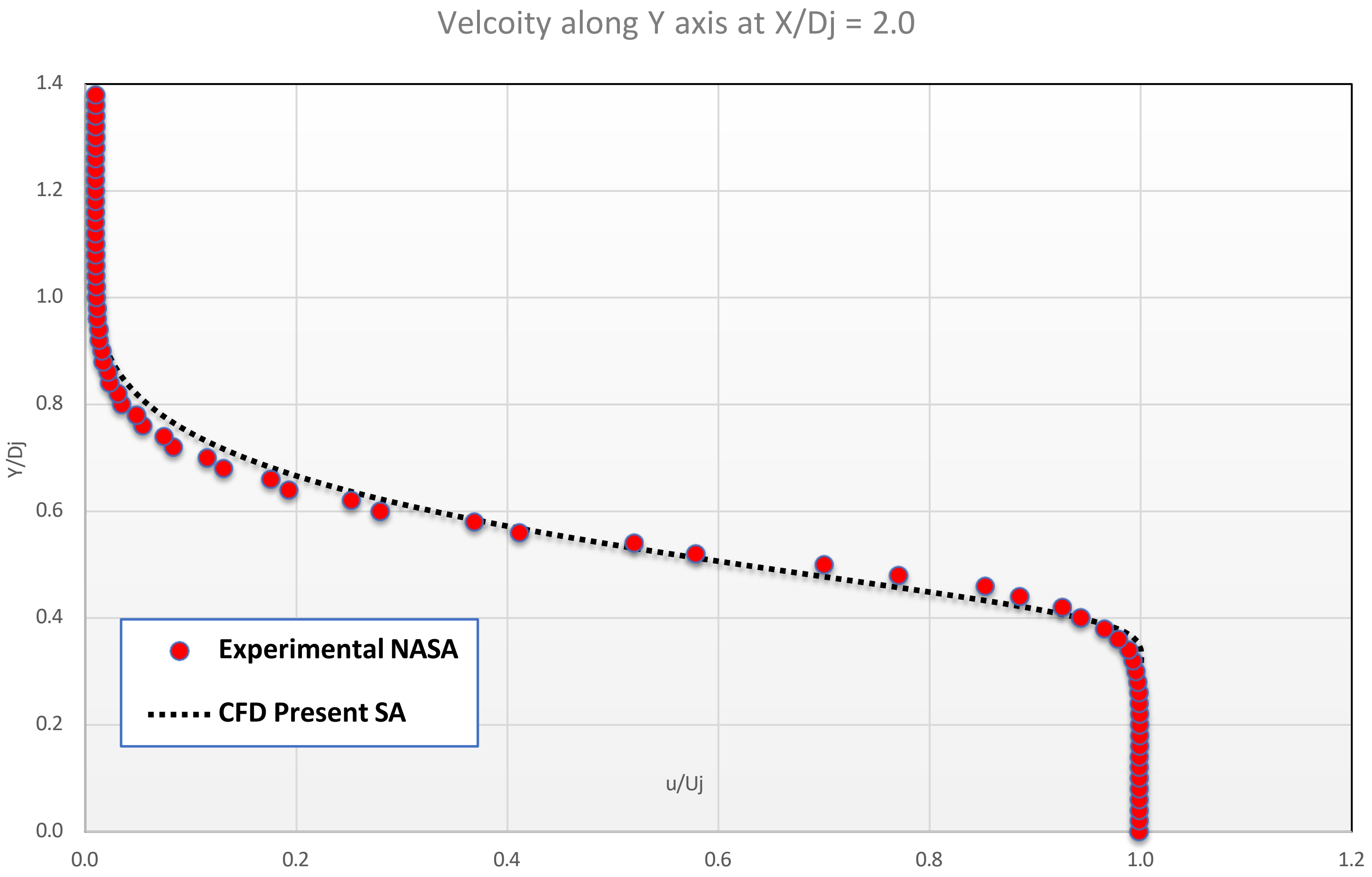
Turbulence Model Predictive Comparison

Jet Core Agreement: Both SA and SST models capture the potential core length and initial jet development accurately up to $X/D_j \approx 5.0$ (errors under 2.2%).

SST Mixing Performance: The SST k-ω model captures shear layer mixing and jet dissipation downstream with a Mean Absolute Percentage Error (MAPE) of 12.06% and a max error of 21.22%.

SA Mixing Limitations: The SA model overpredicts jet decay downstream, underpredicting centerline velocity with an overall MAPE of 27.00% and a max error of 48.57% at $X/D_j = 25.0$.

Validation – Velocity along Y direction at X/Dj = 2



Radial Velocity Validation at X/Dj = 2.0

Potential Core Profile: The normalized radial velocity profile (u/Uj vs Y/Dj) at X/Dj = 2.0 validates the transverse potential core width of the jet.

Shear Layer Validation: The Spalart-Allmaras (SA) model successfully captures the steep velocity gradients within the mixing shear layer boundary (Y/Dj ≈ 0.5).

Momentum Diffusion: Excellent alignment with NASA experimental data confirms that transverse momentum diffusion and mixing layer expansion are correctly modeled.

Conclusion I: Model Validation & Physics

Turbulence Model Validation: Spalart-Allmaras (SA) and SST k-ω models demonstrate excellent accuracy near the nozzle exit ($X/D_j < 5.0$), matching experimental centerline velocity within 2.2% deviation.

Downstream Jet Dissipation: The SST k-ω model shows superior predictive capabilities in the jet decay region ($X/D_j > 10.0$), with an overall profile Mean Absolute Percentage Error (MAPE) of 12.06%.

Velocity Decay Underprediction: The SA model significantly overpredicts jet dissipation further downstream, resulting in a maximum velocity underprediction error of 48.57% at $X/D_j = 25.0$, compared to 21.22% for the SST model.

Conclusion II: Grid Convergence & Setup

SST Grid Convergence: The SST model demonstrates extreme robustness to mesh density, showing a mean velocity difference of only 0.73% between Coarse (70.6k cells) and Fine (214.9k cells) grids.

SA Grid Sensitivity: The SA model exhibits higher grid dependence, showing a 5.13% mean difference between Coarse and Fine grids, which reduces to 1.53% for the Medium grid.

Grid Selection Recommendation: A Medium mesh (128.7k cells) offers an optimal balance of speed and accuracy, achieving convergence within 1.6% relative to the Fine mesh for both turbulence models.

Solver Configuration: The Fluent density-based solver coupled with a second-order momentum discretization scheme successfully resolved the nozzle's compressible flow physics (Mjet = 0.51) and shear layer interactions.